

An Intuitive Geometric Interpretation of the Lagrange Multiplier

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Prerequisite Theory

- Standard Linear Algebra theory
- Differential calculus
- The general concept of manifolds embedded in $\mathbb{R}^n$

The problem

Our motivation in studying this specific problem is to have an easy solution for all problems of the following form:

1) we have a manifold M defined as the pre-image of a regular value of a function g , i.e. $M = g^{-1}(0)$, with 0 (not the scalar, the null vector in $\mathbb{R}^p$) being a regular value of $g : \mathbb{R}^n \rightarrow \mathbb{R}^p$ (dimension of M will be $d = n - p$),

REMINDER: A value c is called a *regular value* of a function g if g is a *submersion* (Dg is surjective) at every point of $g^{-1}(c) = \{x \in \mathbb{R}^n | g(x) = c\}$

2) we have a function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ that is defined on the WHOLE ambient space $\mathbb{R}^n$ and we now want to study the extrema of the function $f = F|_M$, i.e. F with its domain restricted to the manifold M , but

3) we unfortunately cannot directly find a closed formula for Df because it is too difficult, convoluted, impossible to solve, or there simply exists no closed formula.

If we could find a closed formula for Df , the problem would be trivial: find a solution for $Df(x) = 0$ to locate all critical points and then use higher derivatives to determine what kind of extremum it is.

What can we do however without a formula for Df ?

(Note: for the sake of clarity, I will be assuming that all functions are well behaved and derivatives well defined wherever I use them, without explicitly stating it)

The inspiration

We can start dealing with this problem one step at a time. We know we want $Df(x_0)$ to be exactly 0, as in the null matrix, all zeroes, so it should be the case that $Df(x_0)v = 0$ for any vector v , obviously.

But what does ‘any vector v ’ mean in this case? F was defined on the ambient space of $\mathbb{R}^n$ and for F , ‘any vector v ’ would mean any vector v in its domain which is $\mathbb{R}^n$. However, the domain of f is not $\mathbb{R}^n$, it is the manifold M . Therefore we do not care equally about any vector in $\mathbb{R}^n$, but instead only those that the derivative is meaningful to calculate along.

$Df(x_0)v$ essentially means ‘the rate of change of the value of f as we move along direction v ’. In the context of M , we cannot move along any direction; we can only move along directions in its tangent space $T_{x_0}M$!

Therefore to cleanly express this idea once more, we simply want to find a point x_0 where $DF(x_0)v = 0$ for every $v \in T_{x_0}M$!

Building the solution

While our starting idea is totally correct, it does not directly provide us with an easy solution to our original problem, of not being able to compute it. Let’s start by applying subtle transformations and substituting what we already know:

We know that since F is of the form $F : \mathbb{R}^n \rightarrow \mathbb{R}$, Df is a matrix of shape $1 \times n$, i.e. a covector, a linear functional, a transposed vector.

So effectively $DF(x_0)v = 0 \forall v \in T_{x_0}M$, is equivalent to $\nabla F(x_0) \perp v \forall v \in T_{x_0}M$, and since it is perpendicular to every vector in the space, it is perpendicular to the whole space: $\nabla f(x_0) \perp T_{x_0}M$.

This can be rewritten as $\nabla f(x_0) \in (T_{x_0}M)^\perp$ where $(T_{x_0}M)^\perp$ is the perpendicular space of the tangential space of M at x_0 , which is also known as the *normal* space of M , sometimes symbolized as $N_{x_0}M$.

We also know that ∇g_i represent the directions along which the value of g_i change, i.e. the directions along which we would tend to move away from the manifold, the normal space $N_{x_0}M$.

The last key we need to remember is a property we learned in Linear Algebra last semester, that if $v \in \text{span}\{u_i\}$ then “ v can be written as a linear combination of the vectors u_i ”, or “there exists a vector λ such that $v = u\lambda = \sum u_i \lambda_i$ ”. If we write take $u_i = \nabla g_i$, then $N_{x_0}M = \text{span}\{\nabla g_i(x_0)\}$, so $v \in N_{x_0}M \Leftrightarrow v \in \text{span}\{\nabla g_i(x_0)\} \Leftrightarrow \exists \lambda \in \mathbb{R}^p, v = \sum_{i=1}^p \nabla g_i \lambda_i = (Dg(x_0))^T \lambda$.

REMINDER: For any function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\nabla f = (Df)^T$

So our final rewritten version would be:

$$\nabla F(x_0) = \sum_{i=1}^p (\nabla g_i(x_0)) \lambda_i$$

and by transposing both sides, we transform gradients to total derivatives, $\nabla \rightarrow D$, we get:

$$(\nabla F(x_0))^T = \left(\sum_{i=1}^p (\nabla g_i(x_0)) \lambda_i \right)^T$$

$$DF(x_0) = \sum_{i=1}^p \lambda_i (\nabla g_i(x_0))^T$$

$$DF(x_0) = \sum_{i=1}^p \lambda_i (Dg_i(x_0))$$

$$DF(x_0) = \lambda^T Dg(x_0)$$

Which is exactly the result we were looking for!

Putting it all together

To reorganize everything we've seen so far:

- 1) We want to avoid dealing with Df , so we want to rewrite the constraint $Df(x_0) = 0$ using only DF .
- 2) At every *critical point* of f , we need F to have derivative of 0 along any direction that is tangential to our manifold, i.e. $DF(x_0)v = 0 \forall v \in T_{x_0}M$.
- 3) That however just means that the total derivative is perpendicular to the tangent space, so it is part of the *normal* space $DF(x_0) \perp v \forall v \in T_{x_0}M \Rightarrow DF(x_0) \perp T_{x_0}M \Rightarrow DF(x_0) \in N_{x_0}M$.
- 4) The normal space of M at x_0 is simply all the directions along which g tends to change, i.e. the span of ∇g_i since ∇g_i represents the direction of maximum change for each dimension i of g .
- 5) By transforming ∇g_i into Dg_i (transpose) we are now dealing with row vectors instead of column vectors, so we can now replace the gradients with derivatives and say $DF(x_0)$ is a linear combination of the row vectors $Dg_i(x_0)$.
- 6) We can express that linear combination as “existence of a linear functional, a.k.a. transposed vector”, $\exists \lambda \in \mathbb{R}^p$ such that $DF(x_0) = \lambda^T Dg(x_0)$. This particular value of λ , which we can solve for algebraically, is known as the “Lagrange Multiplier”.

Final notes

The constraint that $g(x_0) = 0$ follows naturally from the fact that $x_0 \in M = g^{-1}(0)$.

There is a risk of confusion in such problems due to the fact that we are dealing with both vectors (columns, $n \times 1$) and covectors (rows, $1 \times n$), and often we must use both or switch between them to produce the results we need.

It's important to keep in mind, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$, then:

- $\nabla f(x_0)$ is a vector (column, $n \times 1$)
- $Df(x_0)$ is a covector (row, $1 \times n$)

And the other important difference is that ∇f is only defined for scalar functions ($f : U \rightarrow \mathbb{R}$) while the total derivative Df is also defined for vector-valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^k$ (and is in fact the $k \times n$ Jacobian matrix of f , $J(f)$).

Symbols Used

- $g^{-1}(x_0)$: pre-image of x_0 under g
- x_0 : a specific point of the manifold in question
- Df : total derivative of a function f , equal to its Jacobian
- F : the target function $F : \mathbb{R}^n \rightarrow \mathbb{R}$
- f : the target function with its domain restricted to the manifold ($f = F|_M$)
- g : the function that defines the manifold
- n : the number of dimensions of the ambient space $\mathbb{R}^n$
- p : the number of dimensions of the output of g (the dimensions of the resulting manifold are $n - p$)
- λ : the Lagrange Multiplier, a particular value that represents the linear combination